

PRACTICAL 9 : ENTERPRISE NETWORK DEPLOYMENT AND TROUBLESHOOTING

WEEK 9

Objective

Deploy and troubleshoot a complete Cisco-based enterprise network.

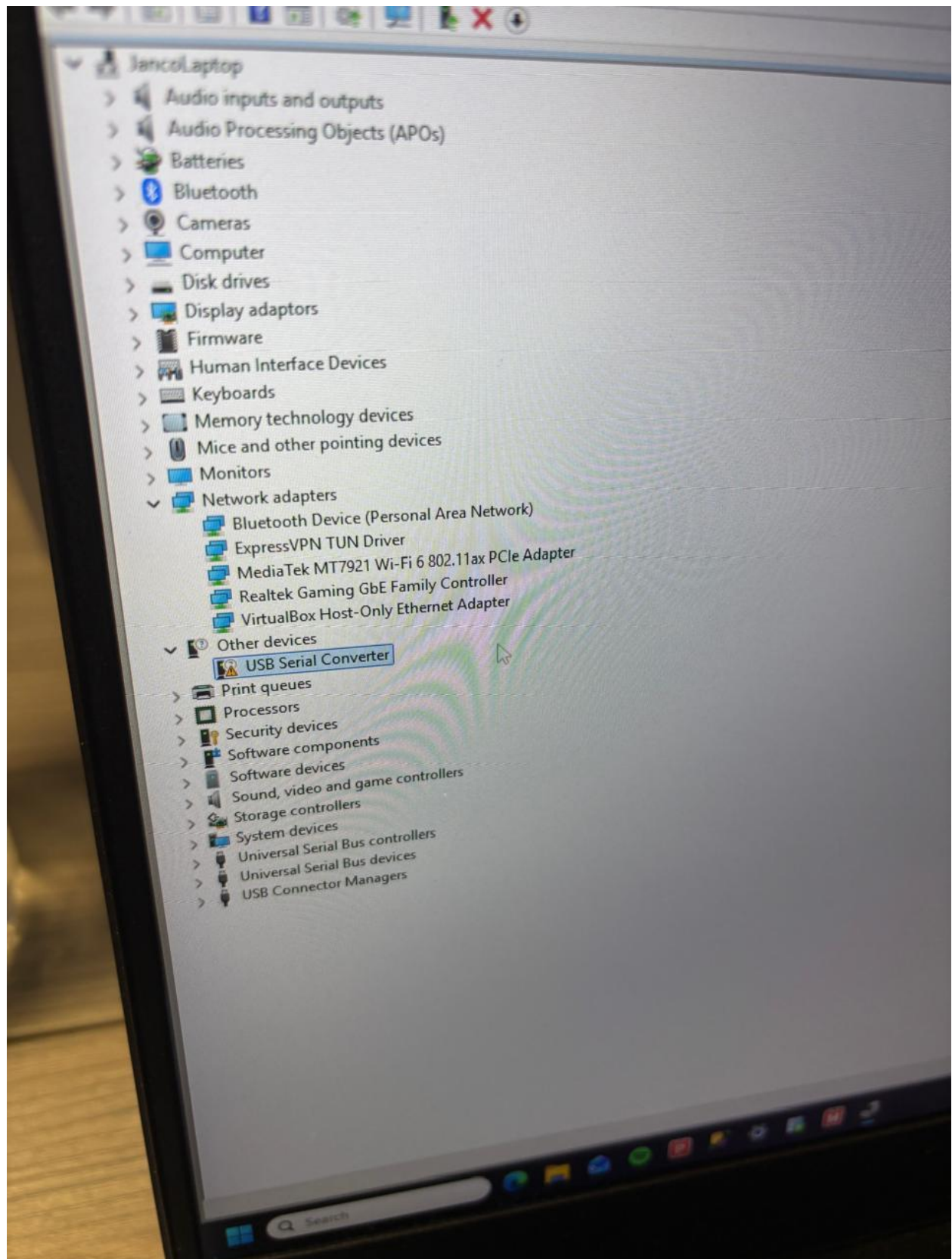
Equipment

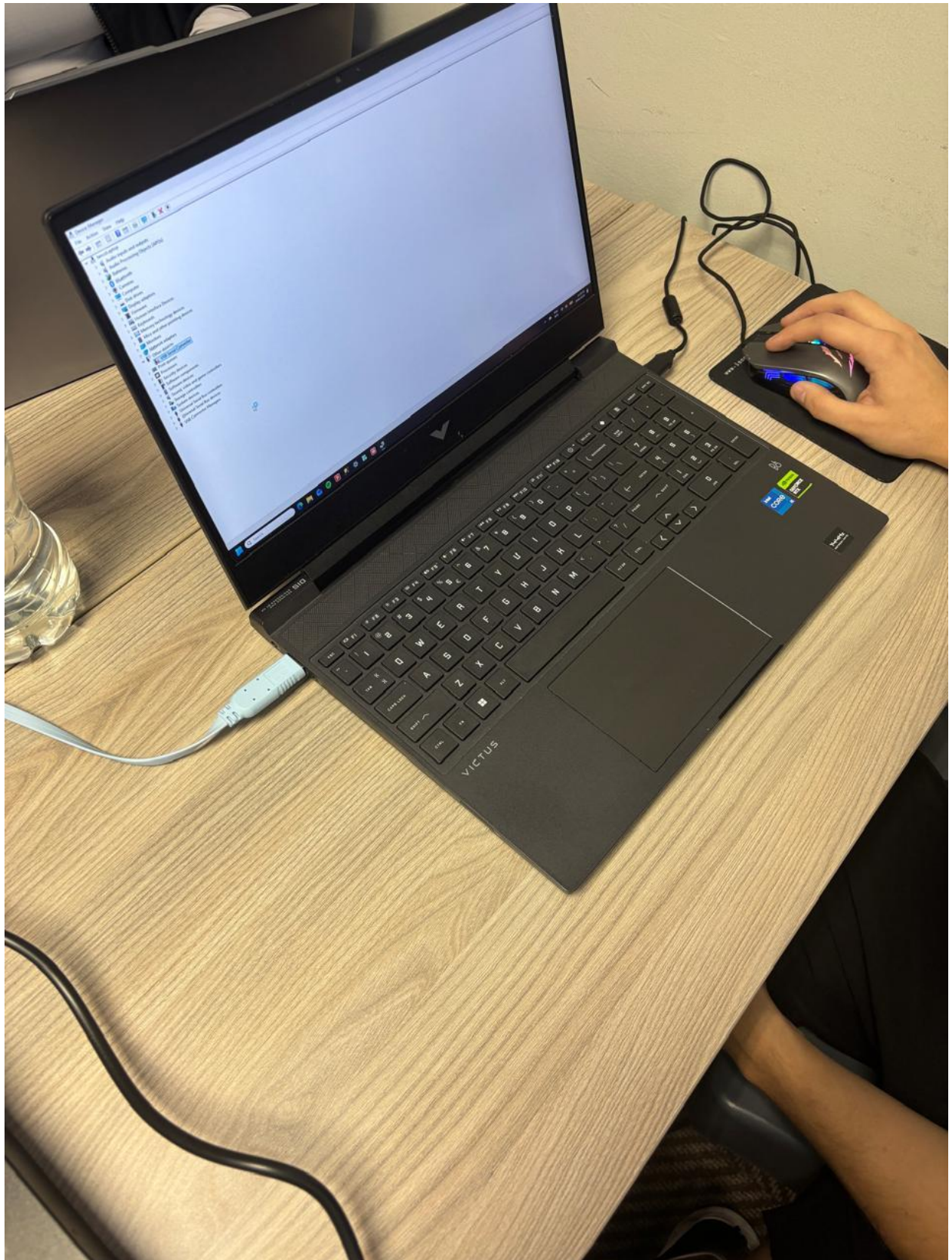
- Cisco Routers
- Cisco Switches
- Switch Cabinets
- Patch Panels

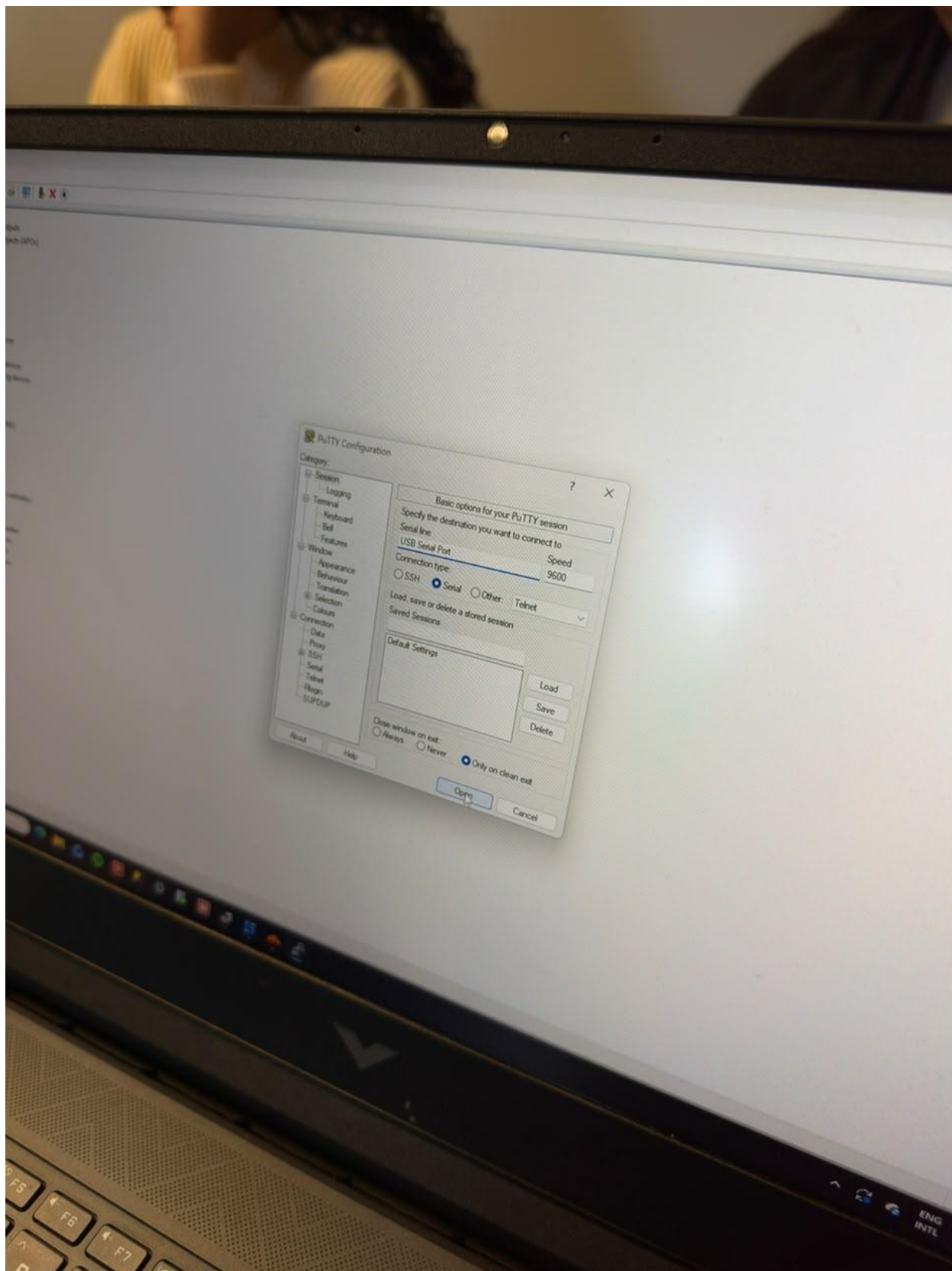
- Printers
- 10 PCs

Activities





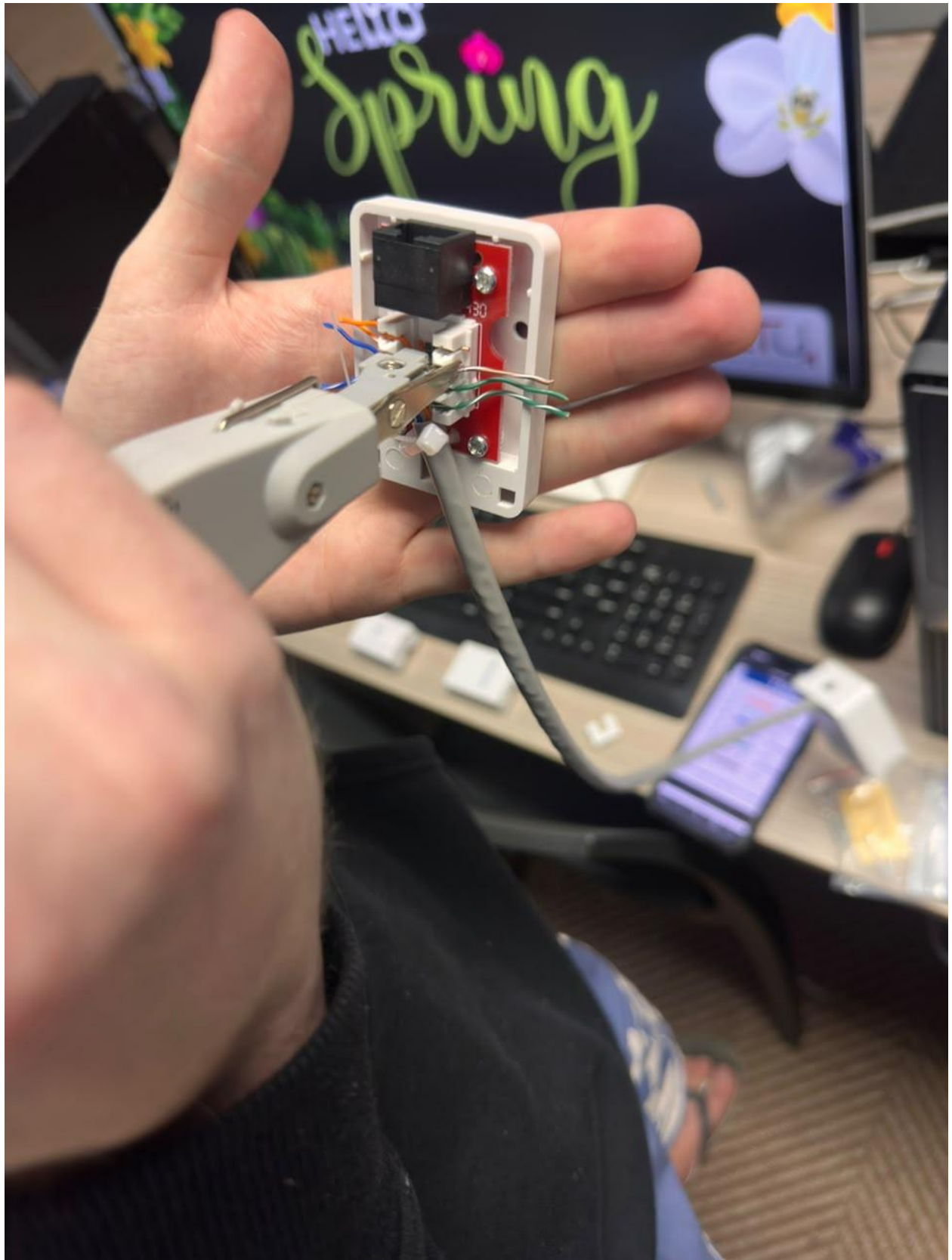






10:57 AM
10/2/2026







PC1 = Fe0/1 ~ 192.168.10.2

PC2 = Fe0/2 ~ 192.168.10.3

PC3 = Fe0/3 ~ 192.168.10.4

PC4 = Fe0/6 ~ 192.168.10.5

PC5 = Fe0/5 ~ 192.168.10.6

Vlan 10

PC6 = Fe0/6 ~ 192.168.20.2

PC7 = Fe0/7 ~ 192.168.20.3

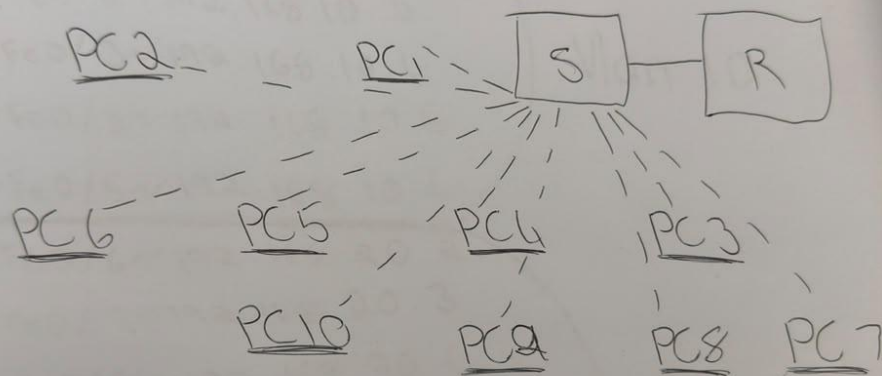
PC8 = Fe0/8 ~ 192.168.20.4

PC9 = Fe0/9 ~ 192.168.20.5

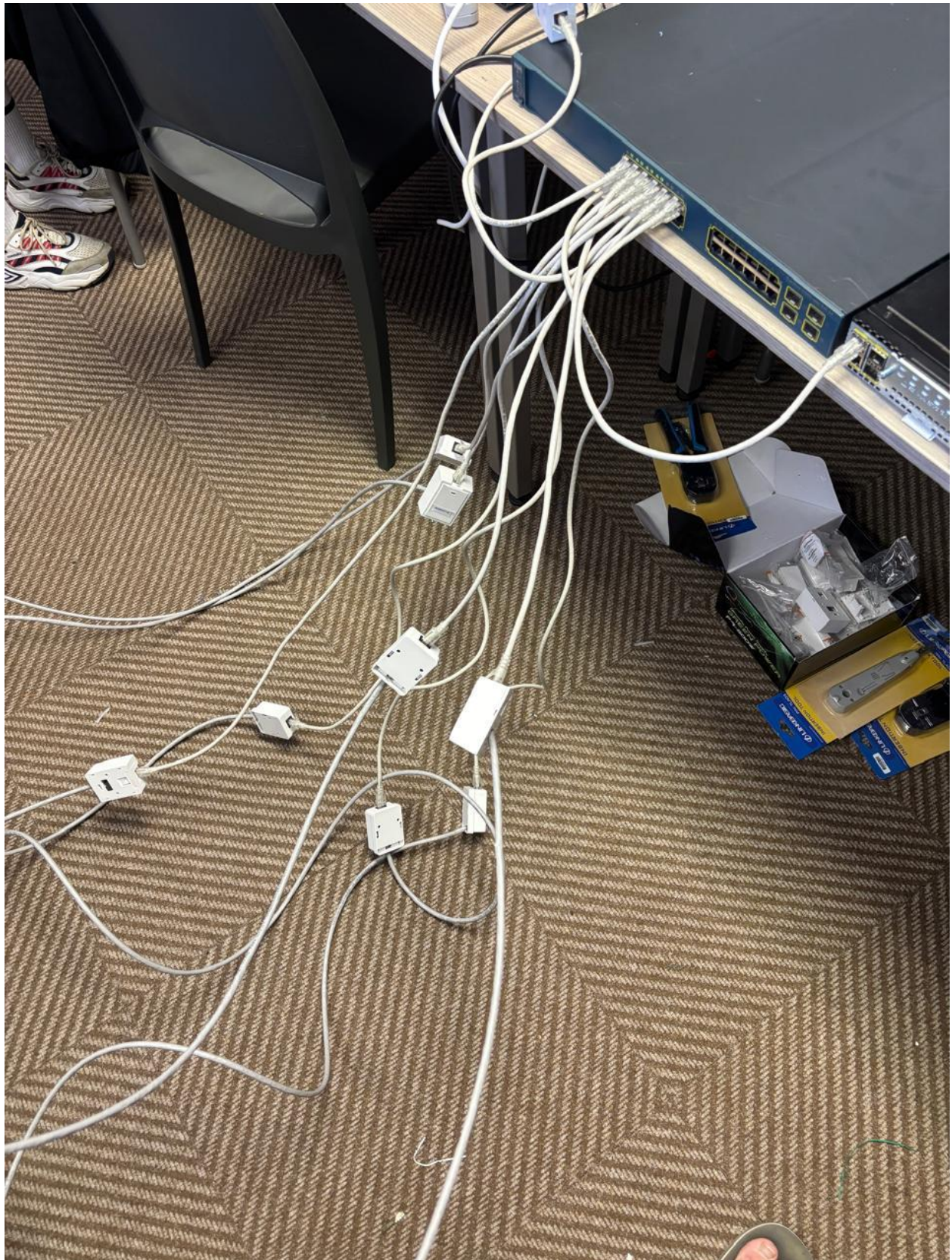
Vlan 20

PC10 = Fe0/10 ~ 192.168.20.6

Topology







1. Install switch cabinet.
2. Install patch panels.
3. Configure switches.
4. Configure routers.
5. Implement VLANs.
6. Configure printers.
7. Test and troubleshoot the network.

Introduced Faults

- Incorrect VLAN assignment
- Incorrect subnet mask
- Incorrect default gateway
- Duplicate IP address
- Faulty patch lead
- Disabled switch port
- Router interface shutdown
- Missing route

Troubleshooting Commands

```
1 show running-config
2 show vlan brief
3 show interfaces
4 show ip route
5 show ip interface brief
6 show mac address-table
7 ping
8 tracert
```

Deliverable

Completed troubleshooting report.

Assessment Rubric (10 Marks)

Criteria	Marks
Fault identification	3
Troubleshooting methodology	3
Correct resolution	2
Professional report	2

Total	10
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PRACTICAL OBSERVATION CHECKLIST

Competency	Yes	No
Uses tools safely	✓	☐
Uses crimping tools correctly	✓	☐
Uses punchdown tools correctly	✓	☐
Applies T568B correctly	✓	☐
Configures Cisco devices correctly	✓	☐
Demonstrates troubleshooting skills	✓	☐
Maintains clean workstation	✓	☐
Demonstrates teamwork	✓	☐
Produces accurate documentation	✓	☐
